

SOUTH KOREA TECHNOLOGY

Korea WFE/Substrate CCL/Chip testing/HBM Supply Chain Check reinforces our positive view on Samsung Electronics, SK Hynix, and SEMCO

In this report, we compile 1) 2Q26 earnings read-across from 20 key non-covered Korean companies across the front-end equipment, substrate/CCL, chip testing, and HBM sectors, 2) takeaways from meetings with 14 of these firms, and 3) additional data-points to summarize current trends in the Korean tech supply chain and provide implications for our coverage universe.

1. Front-end equipment: Korean equipment vendors overall posted solid 2Q26 results and rapidly expanding order backlogs and customer requests for delivery pull-ins reflect an active capex stance from memory players and raise expectations for 2H earnings improvement among equipment suppliers. Strong yoy growth momentum of WFE imports to Korea also supports strong memory WFE demand. This trend aligns with our WFE team's positive industry outlook as well as our views on SEC and Hynix capex.

2. Substrates/CCLs: Korean substrate and CCL suppliers sustained robust earnings growth momentum in 2Q26. Amid tight supply conditions, rapidly expanding order intake raises the likelihood that supply tightness will persist. In line with this, our global team expects the substrate supply shortage to continue for several years. Supported by favorable market conditions and expanding AI revenue, we expect SEMCO substrate revenue to grow significantly this year with meaningful improvement in operating margins.

3. Chip testing: Korean chip testing supply chain companies posted solid 2Q26 results and we believe strong demand for server CPU sockets and a positive demand outlook for memory testing sockets/equipment carry positive implications for our memory coverage.

4. HBMs: Solid 2Q26 earnings, optimistic guidance, and upbeat commentary from HBM supply chain companies carry positive implications for our memory coverage. In line with recent data-points and comments, we estimate the HBM TAM to grow by 67% and 108% yoy this year and next year, respectively.

The authors would like to thank Jiung Kang for his contribution to this report.

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Exhibit 1: Korea tech supply chain overview

WFE Company	BBG Ticker	Market Cap (bn W)	Main Product
Front-end equipment			
Jusung Engineering	036930 KS	7,976	Atomic Layer Deposition (ALD), Chemical Vapor Deposition (CVD) Equipment
Wonik IPS	240810 KS	5,365	PECVD & ALD Systems, Dry Etchers
EO Technics	039030 KS	5,267	Laser Marking, Grooving, Dicing & Annealing Equipment
PSK	319660 KS	3,835	Dry Strip, Dry Cleaning, Wafer Edge Clean
HPSP	403870 KS	3,761	High-Pressure Annealing (HPA) Equipment
Eugene Tech	084370 KS	3,105	Single-Wafer LPCVD, Plasma Treatment Systems
Tes	095610 KS	2,852	PECVD & Gas Phase Etchers
KC Tech	281820 KS	1,350	CMP Equipment & Slurry, Wet Cleaning Systems
SEMES	Private		Clean, Photo, Etch, Test/Package Equipment
Substrate / CCL			
Doosan	000150.KS	19,676	CCL
Isu Petasys	007660 KS	7,657	FCBGA, MLB, FCCSP
Daeduck Electronics	353200 KS	5,643	FC-BGA Substrates, Multilayer PCBs
Simmtech	222800 KS	4,241	Package Substrates (MSAP, FCCSP), Memory Module PCBs
Korea Circuit	007810 KS	1,511	HDI PCBs, Package Substrates (BGA, FC-CSP)
TLB	356860 KS	826	Memory Module PCBs & SSD Module PCBs
Chip testing			
Leeno Industrial	058470 KS	5,274	Spring Contact Probes (LEENO PINs), IC Test Sockets
ISC	095340 KS	3,796	Silicone Rubber Test Sockets
Techwing	089030 KS	1,810	Cube Prober, Probe Station, Memory/SOC Test Handler
Doosan Tesna	131970 KS	1,546	Wafer Probe Testing Services
HBM			
Hanmi Semiconductor	042700 KS	21,493	TC Bonders, Micro SAW & Vision Placement (MSVP)

Market cap as of 2026.08.20

Source: Bloomberg, Company data, Goldman Sachs Global Investment Research

1. Front-end equipment

In this section, we share read-across from 2Q26 earnings results from 9 major Korean front-end equipment makers (Jusung Engineering [Exhibit 17](#), Wonik IPS [Exhibit 18](#), EO Technics [Exhibit 19](#), PSK [Exhibit 20](#), HPSP [Exhibit 21](#), Eugene Tech [Exhibit 22](#), TES [Exhibit 23](#), KC Tech [Exhibit 24](#), SEMES, [Exhibit 25](#)), companies' comments, and data-points. Korean equipment vendors overall posted solid 2Q26 results and rapidly expanding order backlogs and customer requests for delivery pull-ins reflect an active capex stance from memory players and raise expectations for 2H earnings improvement among equipment suppliers. Strong yoy growth momentum of WFE imports to Korea also supports strong memory WFE demand. This trend aligns with our WFE team's positive industry outlook as well as our views on SEC and Hynix capex.

What we learned from 2Q26 results/data-points/company comments:

#1 Driven by solid spending from chip-makers, Korean equipment companies overall posted solid yoy growth in revenue and operating profit in 2Q26 [Exhibit 2](#), extending the yoy improvement trend that began in 1Q26. Notably, a sharp surge in order backlog during the 2Q26 [Exhibit 3](#) indicates a positive signal for 2H26 performance. However, some companies posted sluggish results due to deferred shipments to foundry clients, increased R&D expenses, and one-off cost impacts.

#2 All Korean equipment companies we spoke with said customers are asking to bring forward shipments as DRAM/HBM fab schedules have been moved up and NAND technology upgrade investment is accelerating.

#3 Korean equipment companies clearly stated they are receiving more orders recently, notably since 2Q26. Korean equipment companies are overall optimistic on order flow in the next year, and hence expressed positive tone on 2H26 and 2027 earnings.

#4 Korean equipment companies we spoke with provided revenue growth guidance for this year of +25-40% yoy and commented strong WFE demand is the main driver of growth. In-line with solid revenue growth guidance, we expect SEC/Hynix capex to grow 59%/75% yoy this year [Exhibit 4](#), [Exhibit 5](#), however we expect both DRAM and NAND supply to remain tight [Exhibit 6](#), [Exhibit 7](#).

#5 WFE import value to Korea continued strong yoy growth momentum (up 73% yoy in 2Q26 and up 97% yoy in July 2026) [Exhibit 15](#). Korea's WFE export value to China is showing declining trend (July YTD -20% yoy) [Exhibit 16](#) despite Chinese semiconductor market still growing, which we believe is due to nation-wide focus on localization of semiconductor supply chain.

#6 Korean equipment companies we spoke with commonly mentioned customers' WFE spending is focused on DRAM while overall retained positive tone on NAND WFE outlook as they expect customers' NAND technology upgrade investment to continue.

#7 Majority of Korean equipment companies we spoke with said they expect UTR to continue to increase going forward and are not yet facing capacity tightness as they expanded capacity some years ago. Some companies, however, said they are reviewing plans to add capacity as their UTR is reaching close to full level.

#8 We haven't sensed Korea equipment players are raising equipment pricing, however some suppliers mentioned that they are under discussion or reviewing options to initiate

discussion on normalizing pricing.

#9 Most of the Korean companies we spoke with said there was no notable change in equipment delivery lead times, with some mentioning they are seeing prolonged lead times.

#10 Korean equipment companies commented they are not yet facing tightness in procuring equipment components, however mentioned there could be some impact to display equipment lead times as equipment components companies are prioritizing semiconductor equipment over display equipment.

#11 We expect ongoing memory process node migration <[Exhibit 8](#), [Exhibit 9](#)>, multiple memory fabs currently under construction <[Exhibit 10](#), [Exhibit 11](#), [Exhibit 12](#), [Exhibit 13](#)> and memory makers' large long-term memory fab buildout plan <[Exhibit 14](#)> suggests there are huge growth potentials in WFE TAM in both near-term and long-term, which is in-line with our global WFE team's view where we expect memory WFE capex to grow by 47%/47%/38% yoy in 2026/2027/2028.

#12 EO Technics, which manufactures packaging equipment and PCB equipment as well, commented (1) it saw solid order flow from OSAT companies as one foundry company customer outsourced 2.5D packaging to OSAT companies, and (2) it is receiving strong orders on PCB equipment as PCB equipment supply has become extremely tight, leading to longer lead times from major PCB equipment vendors.

#13 Regarding risk factors, Korean semi equipment vendors that also run display equipment businesses cited weak display demand as a risk factor, while those with China exposure shared a cautious view on long-term order sustainability from China due to the country's equipment localization push.

Implication to our coverage:

#1 Korean equipment companies we spoke with provided solid revenue growth guidance for this year due to strong WFE demand. In-line with solid guidance, we expect SEC/Hynix capex to see strong growth of 59%/75% yoy this year, however expect both DRAM and NAND supply to remain tight despite higher spending for both DRAM and NAND.

#2 We expect ongoing memory process node migration, with multiple memory fabs currently under construction, and memory makers' large long-term memory fab buildout plans suggest there is huge growth potential in WFE TAM in both near-term and long-term, which is in-line with our global WFE team's view where we expect memory WFE capex to grow by 47%/47%/38% yoy in 2026/2027/2028.

Exhibit 2: Combined earnings of 9 Korea major front-end equipment companies

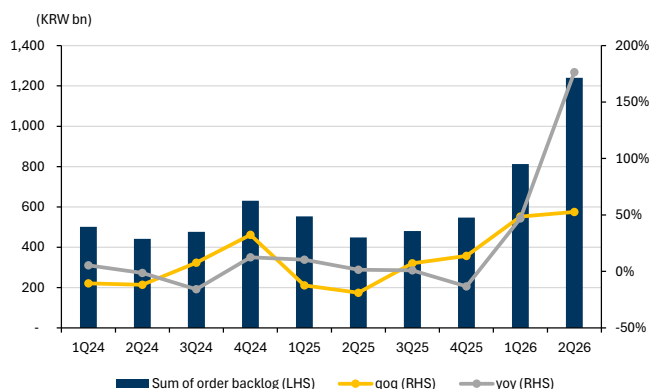
Front-End combined	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue (W mn)	1,139,604	1,332,975	1,359,920	2,884,452	1,251,821	1,427,150	1,360,828	1,747,389	1,474,979	1,716,498	5,048,348	5,466,578	5,787,188
qoq	-18%	17%	2%	112%	-57%	14%	-5%	28%	-16%	16%			
yoy	-16%	6%	29%	108%	10%	7%	0%	-39%	18%	20%	-22%	8%	6%
Gross Profit (W mn)	358,520	424,162	494,920	756,275	451,361	518,512	509,119	572,401	567,117	635,880	1,462,018	1,783,859	2,051,393
GPM (%)	31%	32%	36%	26%	36%	37%	37%	33%	38%	37%	29%	33%	35%
qoq	-8%	18%	17%	53%	-40%	15%	-2%	12%	-1%	12%			
yoy	-10%	12%	69%	93%	26%	22%	3%	-24%	26%	23%	-29%	22%	15%
Operating Profit (W mn)	84,298	142,776	202,581	241,346	162,239	209,340	190,346	187,553	268,747	296,670	306,477	587,630	749,478
OPM (%)	7%	11%	15%	8%	13%	15%	14%	11%	18%	17%	6%	11%	13%
qoq	1%	69%	42%	19%	-33%	29%	-9%	-1%	43%	10%			
yoy	-15%	27%	1742%	190%	92%	47%	-6%	-22%	66%	42%	-65%	92%	28%

Revenue/GP/OP includes Wonik IPS, Eugene Tech, PSK, TES, Jusing Engineering, EO Technics, HPSP, KC Tech, and SEMES

Source: Company filings

Exhibit 3: A sharp surge in order backlog during the 2Q26 indicates a positive signal for 2H26 performance

Sum of order backlog

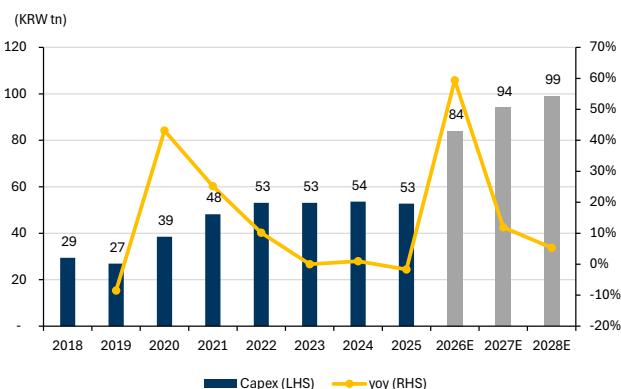


Order backlog of Wonik IPS, Eugene Tech, TES, Jusing Engineering (Semiconductor division), and KC Tech (Semiconductor division) combined

Source: Company filings

Exhibit 4: We expect SEC capex to grow 59% yoy this year

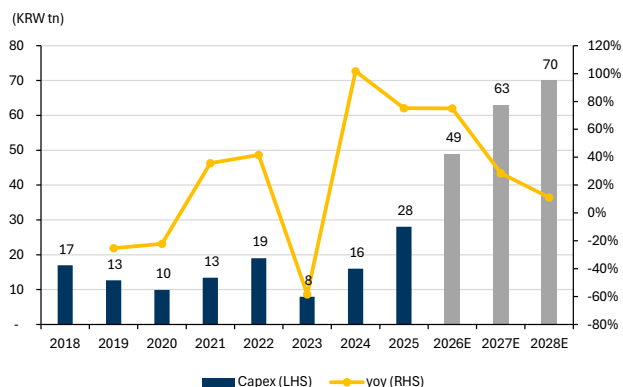
SEC capex trend and estimates



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: We expect Hynix capex to grow 75% yoy this year

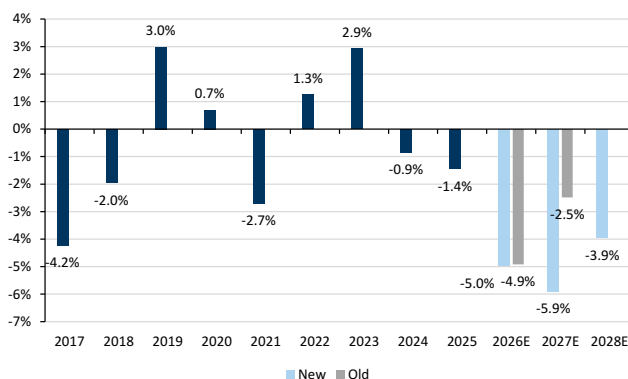
Hynix capex trend and estimates



Source: Company data, Goldman Sachs Global Investment Research

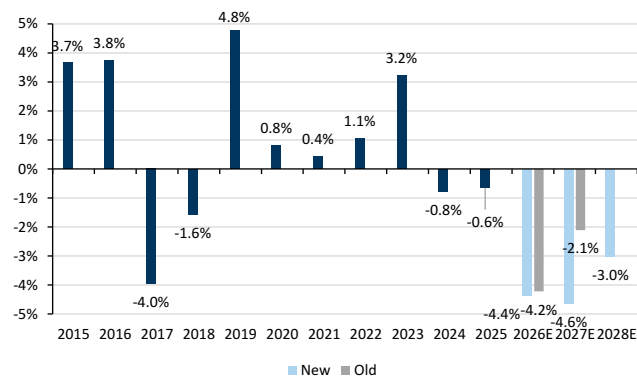
Exhibit 6: We expect 2026E DRAM S/D will be in an undersupply of 5.0%, and 2027E to see a larger undersupply of 5.9%

DRAM S/D



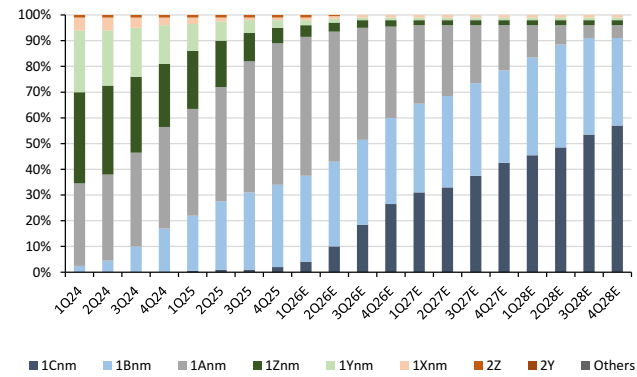
Source: Goldman Sachs Global Investment Research

Exhibit 7: We expect 2026E NAND S/D will be in an undersupply of 4.4%, and 2027E to see undersupply of 4.6%
NAND S/D



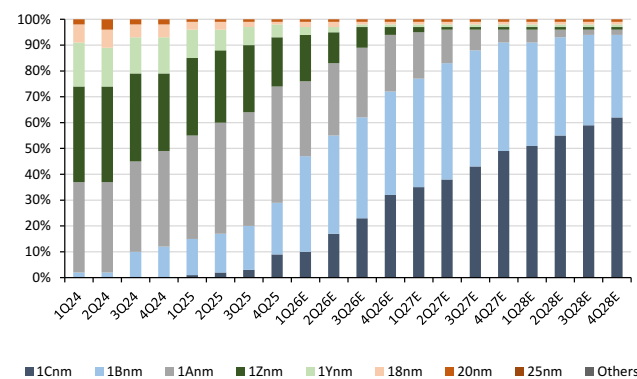
Source: Goldman Sachs Global Investment Research

Exhibit 9: Hynix DRAM node migration estimates



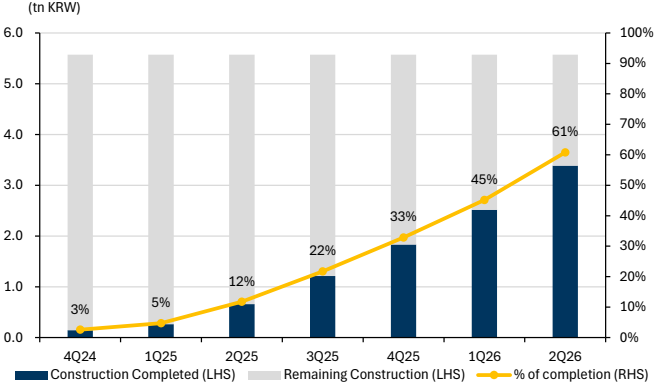
Source: Goldman Sachs Global Investment Research, Company data

Exhibit 8: SEC DRAM node migration estimates



Source: Goldman Sachs Global Investment Research, Company data

Exhibit 10: As of end-2Q26, Hynix Yongin Phase1 fab construction reached 61% progress rate
Yongin Phase 1 fab construction progress based on order delivery rate of SK Ecoplant

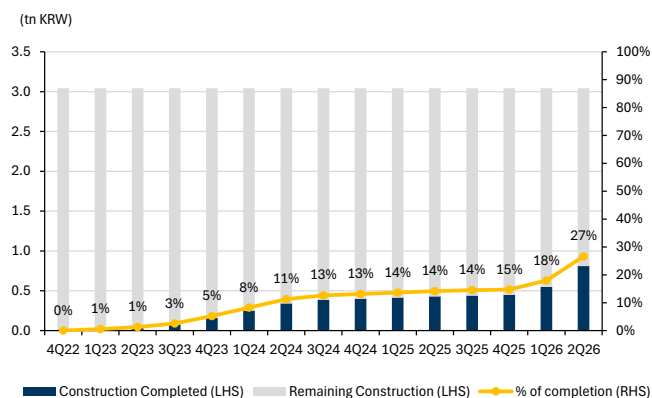


% of completion was calculated by dividing value of construction completed with total construction contract value

Source: Company filings

Exhibit 11: As of end-2Q26, SEC P5 fab construction reached 27% progress rate

P5 fab construction progress based on order delivery rate of Samsung C&T

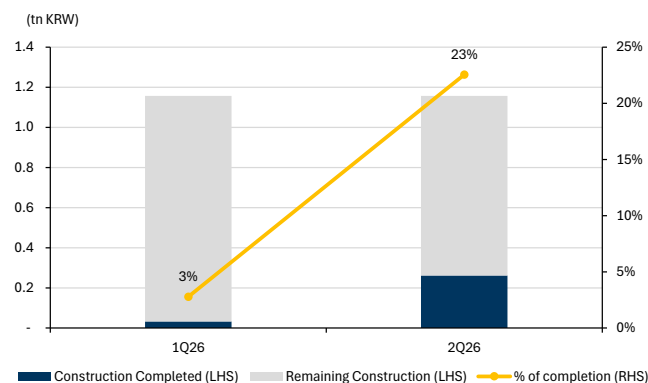


% of completion was calculated by dividing value of construction completed with total construction contract value

Source: Company filings

Exhibit 13: As of end-2Q26, SEC P4 PH2 construction reached 23% progress rate

SEC P4 PH2 construction progress based on Samsung C&T orders

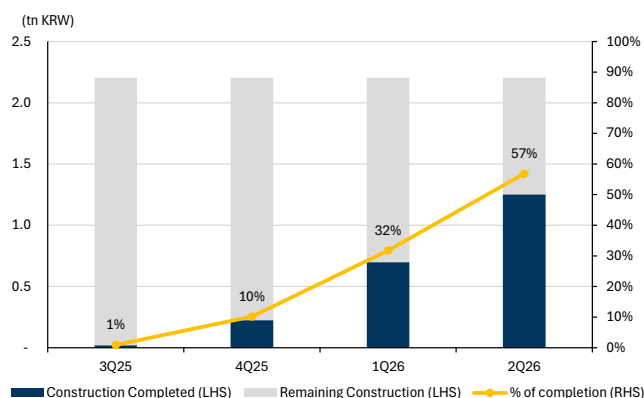


% of completion was calculated by dividing value of construction completed with total construction contract value

Source: Company filings

Exhibit 12: As of end-2Q26, SEC P4 PH4 construction reached 57% progress rate

SEC P4 PH4 construction progress based on Samsung C&T orders



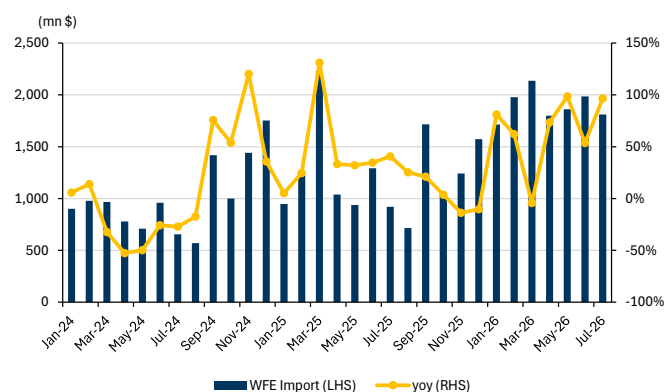
% of completion was calculated by dividing value of construction completed with total construction contract value

Source: Company filings

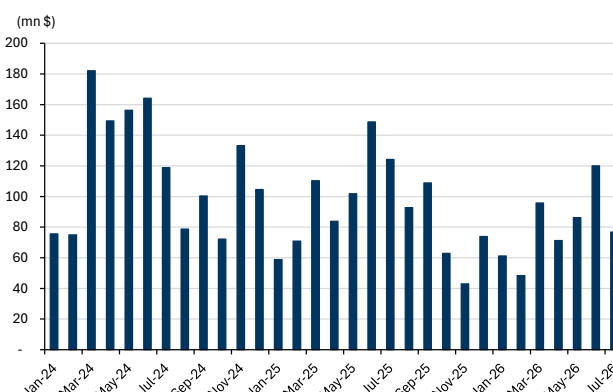
Exhibit 14: Memory makers' long-term fab buildout plan

Company	Fab
SEC	P5 Fab 1
	P6 (P5 Fab 2)
	Gwangju Fab 1
	Gwangju Fab 2
Hynix	Yongin 1
	M17
	Yongin 2
	Yongin 3
	Yongin 4
	Fab 1 in Korea south-west province
Micron	Fab 2 in Korea south-west province
	Idaho 1
	PSMC 1
	PSMC 2
	Idaho 2
	Singapore NAND
	Higashi-Hiroshima
	NY

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 15: Korea WFE import value and yoy trend

Source: TRASS

Exhibit 16: Korea's WFE export value to China showing declining trend (July YTD -20% yoy)
Korea WFE export to China

Source: TRASS

Exhibit 17: Jusung Engineering Financials

Jusung Engineering	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	56,571	97,333	147,205	108,285	120,846	78,792	58,773	52,280	54,879	59,882	284,745	409,393	310,693
qoq	-42%	72%	51%	-26%	12%	-35%	-25%	-11%	5%	9%			
yoy	-18%	207%	71%	10%	114%	-19%	-60%	-52%	-55%	-24%	-35%	44%	-24%
Revenue Breakdown													
Domestic	10,053	10,328	17,665	21,848	34,006	26,351	18,796	30,684	27,214	51,262	39,759	59,894	109,837
Export	46,518	87,005	129,540	86,436	86,841	52,441	39,977	21,597	27,665	8,620	244,986	349,499	200,856
Gross Profit	33,286	67,095	88,101	60,486	72,752	41,960	38,440	24,156	28,627	30,228	150,104	248,967	177,308
GPM (%)	59%	69%	60%	56%	60%	53%	65%	46%	52%	50%	53%	61%	57%
qoq	-34%	102%	31%	-31%	20%	-42%	-8%	-37%	19%	6%			
yoy	-23%	270%	129%	21%	119%	-37%	-56%	-60%	-61%	-28%	-39%	66%	-29%
Operating Profit	7,030	36,063	52,171	1,932	33,908	6,575	3,356	(12,581)	(7,031)	1,423	28,937	97,196	31,258
OPM (%)	12%	37%	35%	2%	28%	8%	6%	-24%	-13%	2%	10%	24%	10%
qoq	-65%	413%	45%	-96%	1655%	-81%	-49%	-475%	-44%	-120%			
yoy	-39%	-513%	744%	-90%	382%	-82%	-94%	-751%	-121%	-78%	-77%	236%	-68%
Order Backlog (Semi)	193,440	161,077	207,311	221,831	140,317	107,627	101,140	88,497	65,371	214,358	230,161	221,831	88,497
qoq	-16%	-17%	29%	7%	-37%	-23%	-6%	-13%	-26%	228%			
yoy	1%	-5%	12%	-4%	-27%	-33%	-51%	-60%	-53%	99%	-3%	-4%	-60%
UTR	63.4%	58.1%	56.5%	110.8%	117.6%	104.0%	103.7%	36.2%	97.5%	98.1%	94.4%	110.8%	36.2%

Source: Company filings

Exhibit 18: Wonik IPS Financials

Wonik IPS	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	113,833	154,902	184,587	294,855	124,198	242,169	268,438	274,991	164,901	216,472	690,338	748,176	909,796
qoq	-50%	36%	19%	60%	-58%	95%	11%	2%	-40%	31%			
yoy	-24%	15%	3%	31%	9%	56%	45%	-7%	33%	-11%	-32%	8%	22%
Gross Profit	41,831	62,783	87,069	100,644	59,972	108,585	106,117	108,170	84,921	98,834	275,930	292,327	382,844
GPM (%)	37%	41%	47%	34%	48%	45%	40%	39%	51%	46%	40%	39%	42%
qoq	-53%	50%	39%	16%	-40%	81%	-2%	2%	-21%	16%			
yoy	-34%	16%	25%	13%	43%	73%	22%	7%	42%	-9%	-33%	6%	31%
Operating Profit	(26,718)	(3,086)	14,466	25,980	(7,381)	36,451	27,539	17,205	10,734	18,383	(18,075)	10,642	73,814
OPM (%)	-23%	-2%	8%	9%	-6%	15%	10%	6%	7%	8%	-3%	1%	8%
qoq	-322%	-88%	-569%	80%	-128%	-594%	-24%	-38%	-38%	71%			
yoy	149%	-83%	-1359%	116%	-72%	-1281%	90%	-34%	-245%	-50%	-119%	-159%	594%
Order Backlog	231,645	160,431	105,167	287,864	260,680	197,749	209,252	298,229	400,388	634,590	245,333	287,864	298,229
qoq	-6%	-31%	-34%	174%	-9%	-24%	6%	43%	34%	58%			
yoy	89%	-14%	-64%	17%	13%	23%	99%	4%	54%	221%	66%	17%	4%

Source: Company filings

Exhibit 19: EO Technics Financials

EO Technics	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	72,840	74,402	82,971	90,733	84,825	94,250	100,514	101,302	115,102	127,799	316,384	320,945	380,891
qoq	2%	2%	12%	9%	-7%	11%	7%	1%	14%	11%			
yoy	-13%	-14%	12%	27%	16%	27%	21%	12%	36%	36%	-29%	1%	19%
Revenue Breakdown													
Domestic	44,672	33,611	41,583	44,549	62,204	18,267	48,585	54,317	48,710	61,192	167,973	164,415	183,373
Export	28,168	40,785	41,417	46,160	22,621	75,983	51,928	46,986	66,392	66,607	148,411	156,530	197,518
Gross Profit	19,735	22,102	24,241	22,947	27,741	37,729	39,536	34,094	44,673	52,760	85,141	89,025	139,099
GPM (%)	27%	30%	29%	25%	33%	40%	39%	34%	39%	41%	27%	28%	37%
qoq	-34%	102%	31%	-31%	20%	-42%	-8%	-37%	19%	6%			
yoy	-23%	270%	129%	21%	119%	-37%	-56%	-60%	-61%	-28%	-39%	66%	-29%
Operating Profit	5,778	6,867	10,433	8,123	14,482	25,825	26,002	14,444	29,798	39,088	28,335	31,201	80,753
OPM (%)	8%	9%	13%	9%	17%	27%	26%	14%	26%	31%	9%	10%	21%
qoq	-65%	413%	45%	-96%	1655%	-81%	-49%	-475%	-44%	-120%			
yoy	-39%	-513%	744%	-90%	382%	-82%	-94%	-751%	-121%	-78%	-77%	236%	-68%

Source: Company filings

Exhibit 20: PSK Financials

PSK	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	77,740	97,133	118,006	105,172	101,959	108,451	104,136	142,667	156,594	167,474	351,864	398,051	457,213
qoq	-28%	25%	21%	-11%	-3%	6%	-4%	37%	10%	7%			
yoy	-7%	48%	26%	-3%	31%	12%	-12%	36%	54%	54%	-24%	13%	15%
Gross Profit	44,563	51,096	60,754	53,181	50,927	51,226	52,024	69,051	77,138	87,186	164,601	209,594	223,228
GPM (%)	57%	53%	51%	51%	50%	47%	50%	48%	49%	52%	47%	53%	49%
qoq	-14%	15%	19%	-12%	-4%	1%	2%	33%	12%	13%			
yoy	28%	88%	20%	2%	14%	0%	-14%	30%	51%	70%	-23%	27%	7%
Operating Profit	21,343	22,782	29,125	10,674	22,634	20,564	23,399	21,898	47,177	50,264	54,119	83,924	88,494
OPM (%)	27%	23%	25%	10%	22%	19%	22%	15%	30%	30%	15%	21%	19%
qoq	34%	7%	28%	-63%	112%	-9%	14%	-6%	115%	7%			
yoy	90%	1449%	14%	-33%	6%	-10%	-20%	105%	108%	144%	-41%	55%	5%
ASP (W mn)	1,294	1,315	1,331	1,453	1,213	1,258	1,358	1,422	1,382	1,382	1,251	1,453	1,422
qoq	3%	2%	1%	9%	-17%	4%	8%	5%	-3%	0%			
yoy	5%	0%	1%	16%	-6%	-4%	2%	-2%	14%	10%	-12%	16%	-2%

Source: Company filings

Exhibit 21: HPSP Financials

HPSP	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	37,753	27,309	49,721	66,618	36,895	51,347	32,047	52,676	31,940	46,123	179,087	181,402	172,965
qoq	24%	-28%	82%	34%	-45%	39%	-38%	64%	-39%	44%			
yoy	-36%	-43%	19%	119%	-2%	88%	-36%	-21%	-13%	-10%	12%	1%	-5%
Gross Profit	26,420	18,780	35,592	48,821	27,011	37,861	22,604	38,814	24,915	33,474	118,589	129,612	126,290
GPM (%)	70%	69%	72%	73%	73%	74%	71%	74%	78%	73%	66%	71%	73%
qoq	37%	-29%	90%	37%	-45%	40%	-40%	72%	-36%	34%			
yoy	-35%	-41%	33%	153%	2%	102%	-36%	-20%	-8%	-12%	8%	9%	-3%
Operating Profit	19,975	12,051	26,164	35,755	18,707	28,644	15,107	27,444	16,136	28,314	95,206	93,946	89,902
OPM (%)	53%	44%	53%	54%	51%	56%	47%	52%	51%	61%	53%	52%	52%
qoq	65%	-40%	117%	37%	-48%	53%	-47%	82%	-41%	75%			
yoy	-43%	-55%	21%	195%	-6%	138%	-42%	-23%	-14%	-1%	12%	-1%	-4%

Source: Company filings

Exhibit 22: Eugene Tech Financials

Eugene Tech	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	68,405	77,716	77,442	114,506	83,154	104,925	62,343	99,903	101,874	122,110	276,533	338,068	350,324
qoq	-3%	14%	0%	48%	-27%	26%	-41%	60%	2%	20%			
yoy	-12%	0%	54%	62%	22%	35%	-19%	-13%	23%	16%	-11%	22%	4%
Gross Profit	29,850	36,583	39,559	59,711	40,975	49,783	32,573	47,618	51,685	59,380	125,651	165,703	170,949
GPM (%)	44%	47%	51%	52%	49%	47%	52%	48%	51%	49%	45%	49%	49%
qoq	-5%	23%	8%	51%	-31%	21%	-35%	46%	9%	15%			
yoy	-14%	4%	62%	90%	37%	36%	-18%	-20%	26%	19%	-17%	32%	3%
Operating Profit	3,971	9,903	15,752	31,551	9,203	19,740	5,182	17,547	18,834	18,221	24,326	61,177	51,672
OPM (%)	6%	13%	20%	28%	11%	19%	8%	18%	18%	15%	9%	18%	15%
qoq	-25%	149%	59%	100%	-71%	115%	-74%	239%	7%	-3%			
yoy	-59%	16%	2092%	493%	132%	99%	-67%	-44%	105%	-8%	-55%	151%	-16%
Order Backlog	48,505	72,566	89,752	84,407	85,329	68,833	76,873	95,646	126,638	153,312	73,007	84,407	95,646
qoq	-34%	50%	24%	-6%	1%	-19%	12%	24%	32%	21%			
yoy	-62%	-17%	13%	16%	76%	-5%	-14%	13%	48%	123%	-53%	16%	13%

Source: Company filings

Exhibit 23: TES Financials

TES	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	42,169	60,513	50,793	86,603	84,496	82,064	67,568	116,987	97,243	123,671	146,938	240,078	351,116
qoq	62%	43%	-16%	71%	-2%	-3%	-18%	73%	-17%	27%			
yoy	16%	6%	83%	233%	100%	36%	33%	35%	15%	51%	-59%	63%	46%
Revenue Breakdown													
DRAM			118,176	66,684	48,163	45,111	37,838	79,557	56,401	76,148	48,490	184,860	210,669
NAND			35,299	19,919	36,333	36,953	29,730	37,430	40,842	47,524	94,040	55,218	140,446
Others			-	-	-	-	-	-	-	-	4,408	-	-
Gross Profit	7,798	15,866	10,320	30,591	22,838	27,202	15,519	26,545	29,685	36,522	12,807	64,574	92,104
GPM (%)	18%	26%	20%	35%	27%	33%	23%	23%	31%	30%	9%	27%	26%
qoq	-242%	103%	-35%	196%	-25%	19%	-43%	71%	12%	23%			
yoy	134%	7%	5310%	-657%	193%	71%	50%	-13%	30%	34%	-85%	404%	43%
Operating Profit	2,386	10,010	4,063	22,017	16,272	20,372	8,416	12,777	22,201	26,027	(5,862)	38,476	57,838
OPM (%)	6%	17%	8%	25%	19%	25%	12%	11%	23%	21%	-4%	16%	16%
qoq	-131%	320%	-59%	442%	-26%	25%	-59%	52%	74%	17%			
yoy	-203%	6%	-176%	-389%	582%	104%	107%	-42%	36%	28%	-110%	-756%	50%
Order Backlog	22,237	46,000	48,835	24,383	55,167	56,313	78,264	41,221	145,538	206,944	9,237	24,383	41,221
qoq	141%	107%	6%	-50%	126%	2%	39%	-47%	253%	42%			
yoy	-38%	739%	326%	164%	148%	22%	60%	69%	164%	267%	-73%	164%	69%

Source: Company filings

Exhibit 24: KC Tech Financials

KC Tech	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	91,602	71,987	76,916	144,929	77,663	97,530	95,964	111,700	156,115	112,689	286,910	385,434	382,857
qoq	28%	-21%	7%	88%	-46%	26%	-2%	16%	40%	-28%			
yoy	46%	5%	-9%	103%	-15%	35%	25%	-23%	101%	16%	-24%	34%	-1%
Revenue Breakdown													
Domestic	68,720	56,019	60,325	127,833	57,094	67,131	84,961	98,868	139,325	99,434	212,495	312,897	308,053
China	20,322	11,243	12,865	13,688	17,841	25,840	6,588	8,751	13,850	7,705	51,047	58,118	59,021
US	2,538	4,571	3,723	3,408	2,614	4,559	4,264	4,080	2,934	5,544	22,353	14,240	15,516
Others	22	154	2	-	114	-	151	2	6	6	1,015	178	266
Gross Profit	28,192	26,685	27,247	41,668	27,096	37,016	35,846	42,538	59,102	46,570	98,368	123,792	142,496
GPM (%)	31%	37%	35%	29%	35%	38%	37%	38%	38%	41%	34%	32%	37%
qoq	5%	-5%	2%	53%	-35%	37%	-3%	19%	39%	-21%			
yoy	58%	12%	-8%	55%	-4%	39%	32%	2%	118%	26%	-25%	26%	15%
Operating Profit	10,016	5,330	9,024	25,413	7,835	15,558	14,868	21,704	34,811	23,522	32,704	49,783	59,966
OPM (%)	11%	7%	12%	18%	10%	16%	15%	19%	22%	21%	11%	13%	16%
qoq	-8%	-47%	69%	182%	-69%	99%	-4%	46%	60%	-32%			
yoy	2988%	-37%	-31%	134%	-22%	192%	65%	-15%	344%	51%	-46%	52%	20%
Order Backlog	39,052	31,272	42,728	18,268	18,554	23,868	21,548	37,532	86,466	34,501	40,756	18,268	37,532
Semiconductor	5,226	1,742	25,215	11,749	11,415	18,123	15,173	23,431	74,535	31,443	2,399	11,749	23,431
Display	33,826	29,531	17,513	6,519	7,139	5,744	6,375	14,101	11,931	3,058	38,357	6,519	14,101

Source: Company filings

Exhibit 25: SEMES Financials

SEMES	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	578,691	671,681	572,279	####	537,785	567,622	571,045	794,882	596,332	740,277	2,515,549	2,445,031	2,471,335
qoq	-15%	16%	-15%	227%	-71%	6%	1%	39%	-25%	24%			
yoy	-21%	-3%	37%	175%	-7%	-15%	0%	-58%	11%	30%	-13%	-3%	1%
Gross Profit	126,845	123,173	122,038	338,227	122,050	127,150	166,460	181,414	166,370	190,927	430,827	460,265	597,074
GPM (%)	22%	18%	21%	18%	23%	22%	29%	23%	28%	26%	17%	19%	24%
qoq	15%	-3%	-1%	177%	-64%	4%	31%	9%	-8%	15%			
yoy	-8%	-16%	240%	206%	-4%	3%	36%	-46%	36%	50%	-24%	7%	30%
Operating Profit	40,517	42,855	41,383	79,902	46,579	35,610	66,476	67,116	96,087	91,428	66,788	121,284	215,781
OPM (%)	7%	6%	7%	4%	9%	6%	12%	8%	16%	12%	3%	5%	9%
qoq	310%	6%	-3%	93%	-42%	-24%	87%	1%	43%	-5%			
yoy	15%	-42%	-179%	708%	15%	-17%	61%	-16%	106%	157%	-70%	82%	78%

Source: Company filings

2. Substrate/CCL

In this section, we share read-across from 2Q26 earnings results of 6 major Korean substrate/CCL makers (Doosan BG [Exhibit 32](#), Isu Petasys (Parent) [Exhibit 33](#), Daeduck Electronics [Exhibit 34](#), Simmtech [Exhibit 35](#), Korea Circuit [Exhibit 36](#), TLB [Exhibit 37](#)), companies' comments, and data-points. Korean substrate and CCL suppliers sustained robust earnings growth momentum in 2Q26. Amid tight supply conditions, rapidly expanding order intake raises the likelihood that supply tightness will persist. In line with this, our global team expects the substrate supply shortage to continue for several years. Supported by favorable market conditions and expanding AI revenue, we expect SEMCO substrate revenue to grow significantly this year with meaningful improvement in operating margins.

What we learned from 2Q26 results/data-points/company comments:

- #1** Driven by increased shipment volumes amid solid end-demand and ASP improvements from an expanded high-value product mix, Korean substrate and CCL makers posted strong 2Q26 results, sustaining robust yoy growth momentum <[Exhibit 26](#)>. Notably, order backlogs in 2Q26 for substrate companies we track all saw triple digit yoy growth implying strong sustainability in earnings growth <[Exhibit 27](#)>.
- #2** Companies we spoke with mentioned that overall supply of substrates/CCLs is tight with increasing utilization rate due to strong orders, hence most of the companies we spoke with are either undergoing capacity addition or reviewing new capacity adds. This is in-line with our global team's view where we expect ABF substrate S/D gap to widen from an 8% undersupply in 2026 to 34/51% in 2027/28 <[Exhibit 28](#)>.
- #3** Due to tightening supply and rising raw material costs, a number of substrate/CCL companies raised pricing between March and June, and some mentioned they are planning to raise pricing for additional products as well.
- #4** Korean substrate/CCL companies we spoke with expect revenue to grow 40-50% yoy this year and expect significant margin improvement. Notably, ISU Petasys expects strong earnings to continue for next year as the company plans to scale up its multi-lam production in 2H26. Korean substrate/CCL companies' guidance is in-line with our view on SEMCO, where we expect its substrate business to see 51% yoy revenue growth this year and OPM improvement by 10%p from 6% to 16% this year.
- #5** Some companies mentioned that lead times have extended due to strong demand and tight raw material supply (T-glass was frequently mentioned), but noted this is not yet at a level that would pose a risk to earnings.
- #6** Companies we spoke with said overall demand is solid across major applications. In particular, they noted strong demand for AI accelerators, 800G switches, automotive, memory (GDDR7, SSD controllers), and optical modules.
- #7** Simmtech, main supplier of module PCB and MCP substrates for SOCAMM modules, said that SOCAMM-related demand is rising faster than their initial expectations, and raised its SOCAMM-related revenue guidance for this year by over 50% vs. beginning of this year, and expect meaningful SOCAMM revenue contribution this year. In-line with this, we expect SOCAMM demand to increase by 420% yoy this year and reach 7% of global DRAM demand this year <[Exhibit 31](#)>.
- #8** Korean substrate/CCL companies we spoke with hold an overall positive view on the industry outlook, and cited tight raw material procurement and revenue concentration on specific customers as potential risk factors.

Implication to our coverage:

- #1** Korean substrate/CCL companies are saying overall supply is tight and hold overall positive view on the industry outlook, which is in-line with our global team's view where we expect ABF substrate S/D gap to widen from an 8% undersupply in 2026 to 34/51% in 2027/28.
- #2** Korean substrate/CCL companies we spoke with expect revenue to grow 40-50% yoy this year and expect significant margin improvement, which is in-line with our view on SEMCO where we expect its substrate business to see 51% yoy revenue growth this year

and OPM improvement by 10%p from 6% to 16% this year.

#3 Simmtech is seeing stronger-than-expected SOCAMM-related demand, hence raised its SOCAMM-related revenue guidance for this year by over 50% vs. beginning of this year, and expect meaningful SOCAMM revenue contribution this year. In-line with this, we expect SOCAMM demand to increase by 420% yoy this year and reach 7% of global DRAM demand this year.

Exhibit 26: Combined earnings of 6 Korea major Substrate / CCL companies

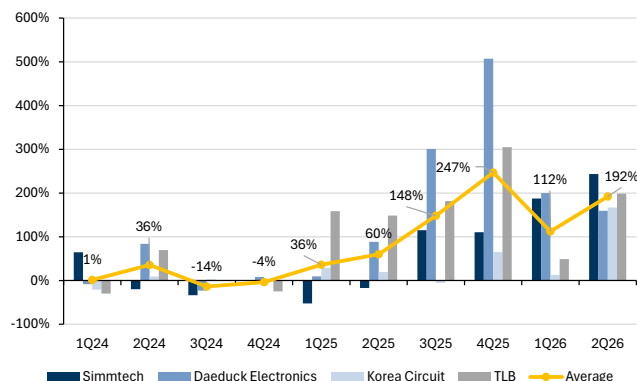
Substrate / CCL combined	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue (W mn)	1,262,655	1,374,250	1,376,384	1,423,153	1,539,023	1,667,934	1,801,748	2,040,771	2,158,114	2,395,215	4,848,189	5,436,442	7,049,476
qoq	1%	9%	0%	3%	8%	8%	8%	13%	6%	11%			
yoy	14%	14%	7%	14%	22%	21%	31%	43%	40%	44%	-23%	12%	30%
Operating Profit (W mn)	6,153	64,387	52,802	12,213	127,141	188,394	206,741	255,526	333,680	421,405	(8,590)	135,554	777,802
OPM (%)	0%	5%	4%	1%	8%	11%	11%	13%	15%	18%	0%	2%	11%
qoq	-126%	946%	-18%	-77%	941%	48%	10%	24%	31%	26%			
yoy	-141%	485%	182%	-153%	1966%	193%	292%	1992%	162%	124%	-101%	-1678%	474%

Revenue/OP includes Doosan BG, Simmtech, Daeduck Electronics, Korea Circuit, TLB, and Isu Petasys (Parent)

Source: Company filings

Exhibit 27: Order backlog in 2Q26 for substrate companies we track all saw triple digit yoy growth

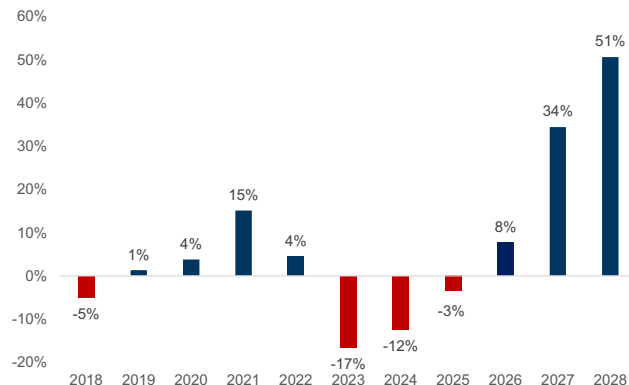
Order backlog growth rate of substrate companies



Source: Company filings

Exhibit 28: We expect the ABF industry S/D gap to continue expanding in coming years

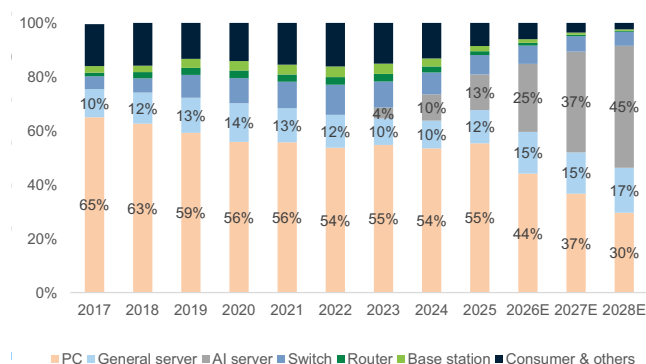
ABF supply shortage ratio



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 29: AI server contribution continues to increase

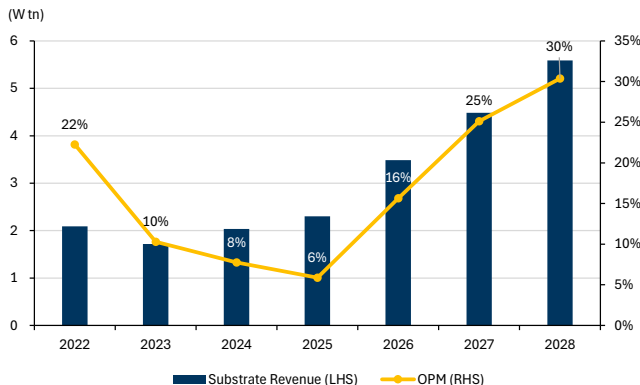
ABF demand by application



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 30: We expect SEMCO's substrate business to see 51% yoy revenue growth this year and OPM improvement by 10%p

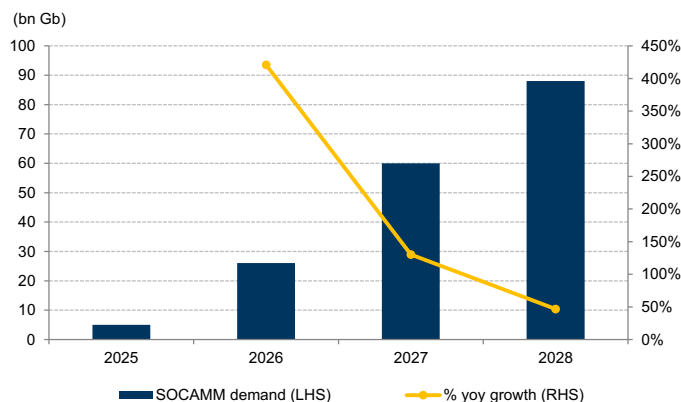
SEMCO substrate revenue and OPM



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 31: We expect SOCAMM demand to increase by 420% yoy this year and reach 7% of global DRAM demand this year

SOCAMM demand and yoy growth



Source: Goldman Sachs Global Investment Research

Exhibit 32: Doosan BG Financials

Doosan BG	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	186,546	235,219	249,556	335,877	402,869	476,202	439,922	556,741	617,345	676,007	814,055	1,007,198	1,875,734
qoq	-12%	26%	6%	35%	20%	18%	-8%	27%	11%	10%	-14%	24%	86%
yoy	10%	12%	13%	58%	116%	102%	76%	66%	53%	42%	4%	12%	26%
Operating Profit	16,832	31,272	29,698	44,778	116,109	136,204	103,895	128,746	185,643	201,270	34,642	122,580	484,954
OPM (%)	9%	13%	12%	13%	29%	29%	24%	23%	30%	30%	4%	12%	26%
qoq	-744%	86%	-5%	51%	159%	17%	-24%	24%	44%	8%	-60%	254%	296%
yoy	250%	83%	94%	-1812%	590%	336%	250%	188%	60%	48%	-216%	-297%	274%
Net Income	11,542	27,088	20,055	40,562	90,430	91,406	88,338	100,836	155,258	162,095	(50,348)	99,247	371,010
NPM (%)	6%	12%	8%	12%	22%	19%	20%	18%	25%	24%	-6%	10%	20%
qoq	-400%	135%	-26%	102%	123%	1%	-3%	14%	54%	4%	-216%	-297%	274%
yoy	568%	-143%	36%	-1154%	683%	237%	340%	149%	72%	77%	47,308	51,022	60,127
CCL ASP (KRW)	50,154	50,298	49,676	51,022	55,316	58,794	58,855	60,127	68,799	69,132	-11%	8%	18%
qoq	6%	0%	-1%	3%	8%	6%	0%	2%	14%	0%	-11%	8%	18%
yoy	16%	6%	6%	8%	10%	17%	18%	18%	24%	18%	-11%	8%	18%

Source: Company filings

Exhibit 33: ISU Petasys (Parent) Financials

ISU Petasys (Parent)	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	168,365	174,197	181,599	194,568	209,604	209,706	258,274	252,082	276,348	308,245	579,045	718,729	929,665
qoq	14%	3%	4%	7%	8%	0%	23%	-2%	10%	12%	16%	24%	29%
yoy	19%	21%	25%	31%	24%	20%	42%	30%	32%	47%	16%	24%	29%
Revenue Breakdown													
Datacenter			43,584	46,696	155,107	163,570	165,295	168,895	196,207	221,936	295,313	531,859	652,867
Network			138,015	147,872	54,497	46,135	92,979	83,187	80,141	86,309	283,732	186,870	276,798
Gross Profit	27,612	33,754	30,991	31,762	50,947	45,164	59,346	65,806	59,861	72,176	90,803	124,120	221,263
GPM (%)	16%	19%	17%	16%	24%	22%	23%	26%	22%	23%	16%	17%	24%
qoq	21%	22%	-8%	2%	60%	-11%	31%	11%	-9%	21%	-30%	37%	78%
yoy	12%	45%	54%	39%	85%	34%	91%	107%	17%	60%	50,217	78,480	152,139
Operating Profit	16,467	23,257	20,238	18,518	33,350	31,767	44,092	42,929	45,866	54,906	9%	11%	16%
OPM (%)	10%	13%	11%	10%	16%	15%	17%	17%	17%	18%	9%	11%	16%
qoq	73%	41%	-13%	-8%	80%	-5%	39%	-3%	7%	20%	-48%	56%	94%
yoy	9%	47%	107%	95%	103%	37%	118%	132%	38%	73%	3.0	3.7	4.5
ASP (W mn per m ²)	3.4	3.6	3.6	3.7	4.5	4.4	4.5	4.5	4.6	4.9	2.0	2.5	2.2
Domestic	2.0	2.1	2.7	2.5	1.9	2.1	2.1	2.2			3.1	3.7	4.5
Export	3.4	3.6	3.6	3.7	4.6	4.4	4.5	4.5	4.6	4.9	64%	71%	69%
Order Backlog Breakdown											36%	29%	31%
Datacenter	49%	65%	70%	73%	68%	78%	70%	69%	67%				
Network	51%	35%	30%	27%	32%	22%	30%	31%	33%				

Source: Company filings

Exhibit 34: Daeduck Electronics Financials

Daeduck Electronics	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	214,766	238,239	232,654	206,477	215,376	245,858	286,173	317,888	346,314	400,975	909,651	892,136	1,065,295
qoq	-8%	11%	-2%	-11%	4%	14%	16%	11%	9%	16%			
yoy	-1%	8%	-2%	-12%	0%	3%	23%	54%	61%	63%	-31%	-2%	19%
Revenue Breakdown													
FCCSP							41%	41%	39%	43%			
CSP							27%	23%	22%	20%			
FCBGA							17%	22%	23%	23%			
MLB							15%	14%	16%	14%			
Gross Profit	8,800	23,916	21,564	7,267	5,646	17,009	39,580	46,055	72,401	107,643	75,135	61,547	108,290
GPM (%)	4%	10%	9%	4%	3%	7%	14%	14%	21%	27%	8%	7%	10%
qoq	-55%	172%	-10%	-66%	-22%	201%	133%	16%	57%	49%			
yoy	-62%	28%	58%	-63%	-36%	-29%	84%	534%	1182%	533%	-75%	-18%	76%
Operating Profit	(2,886)	10,904	9,218	(5,976)	(6,202)	1,866	24,448	28,949	51,298	70,276	23,732	11,259	49,061
OPM (%)	-1%	5%	4%	-3%	-3%	1%	9%	9%	15%	18%	3%	1%	5%
qoq	-145%	-478%	-15%	-165%	4%	-130%	1210%	18%	77%	37%			
yoy	-128%	94%	551%	-193%	115%	-83%	165%	-584%	-927%	3666%	-90%	-53%	336%
PCB ASP (W mn)													
Domestic	1.66	1.80	2.05	2.24	2.62	2.51	2.39	2.44	2.71	2.66	1.64	2.24	2.44
Export	1.20	1.21	1.24	1.26	1.48	1.42	1.39	1.44	1.86	1.95	1.31	1.26	1.44
Order Backlog	72,300	68,200	77,600	35,700	79,200	128,300	311,000	216,800	237,400	332,800	33,100	35,700	216,800
qoq	118%	-6%	14%	-54%	122%	62%	142%	-30%	10%	40%			
yoy	-9%	84%	-23%	8%	10%	88%	301%	507%	200%	159%	4%	8%	507%

Source: Company filings

Exhibit 35: Simmtech Financials

Simmtech	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	293,867	311,661	324,781	301,112	303,597	340,767	372,839	393,356	422,371	514,565	1,041,896	1,231,421	1,410,560
qoq	-1%	6%	4%	-7%	1%	12%	9%	6%	7%	22%			
yoy	44%	23%	13%	1%	3%	9%	15%	31%	39%	51%	-39%	18%	15%
Revenue Breakdown													
MSAP	182,400	196,100	202,300	168,500	170,800	196,300	228,800	260,000	260,500	307,200	685,000	749,200	855,900
HDI	73,700	80,700	90,700	81,500	88,500	92,200	88,200	77,900	97,400	127,800	113,000	326,600	346,700
Tenting	37,800	34,800	31,800	53,200	44,300	52,300	55,800	55,500	64,400	79,600	244,000	157,700	207,900
Gross Profit	10,130	27,339	20,879	(6,400)	10,218	36,202	41,027	35,945	43,876	95,153	28,446	51,948	123,391
GPM (%)	3%	9%	6%	-2%	3%	11%	11%	9%	10%	18%	3%	4%	9%
qoq	-1053%	170%	-24%	-131%	-260%	254%	13%	-12%	22%	117%			
yoy	-1058%	204%	-3%	502%	1%	32%	96%	-662%	329%	163%		83%	138%
Operating Profit	(14,934)	3,610	510	(36,156)	(16,324)	5,544	12,389	10,260	13,711	62,890	(88,124)	(46,969)	11,869
OPM (%)	-5%	1%	0%	-12%	-5%	2%	3%	3%	3%	12%	-8%	-4%	1%
qoq	-48%	-124%	-86%	-7187%	-55%	-134%	123%	-17%	34%	359%			
yoy	-54%	-117%	-109%	26%	9%	54%	2328%	-128%	-184%	1034%	-125%	-47%	-125%
PCB ASP (W mn per m ²)	1.07	1.08	1.08	1.08	1.05	1.09	1.10	1.13	1.34	1.46	1.07	1.08	1.13
qoq	0%	0%	0%	1%	-3%	4%	1%	2%	19%	9%			
yoy	-2%	-1%	0%	1%	-2%	1%	2%	4%	28%	34%	-1%	1%	4%
Order Backlog	367,531	193,051	171,114	202,132	174,333	160,062	367,417	425,483	501,421	549,297	202,201	202,132	425,483
qoq	82%	-47%	-11%	18%	-14%	-8%	130%	16%	18%	10%			
yoy	64%	-20%	-34%	0%	-53%	-17%	115%	110%	188%	243%	-27%	0%	110%

Assumed 2Q26 Order Backlog fx rate as 1,400 KRW per USD

Source: Company filings

Exhibit 36: Korea Circuit Financials

Korea Circuit	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	354,863	375,451	341,348	335,295	354,588	331,348	375,652	448,138	419,901	407,180	1,332,234	1,406,957	1,509,725
qoq	14%	6%	-9%	-2%	6%	-7%	13%	19%	-6%	-3%			
yoy	5%	13%	-3%	7%	0%	-12%	10%	34%	18%	23%	-17%	6%	7%
Revenue Breakdown													
PCB	186,606	198,228	199,304	204,243	196,673	219,765	247,694	273,807	263,546	281,932	706,074	788,381	937,939
FPCB	131,294	142,360	113,644	101,984	129,582	86,253	101,089	147,537	128,126	90,109	430,069	489,282	464,461
Packaging	34,507	31,475	25,659	26,510	25,057	26,540	25,758	25,692	26,533	33,514	185,485	118,151	103,047
Others	2,456	3,389	2,740	2,558	3,276	(1,210)	1,110	1,102	1,696	1,624	10,607	11,143	4,278
Gross Profit	10,045	17,832	5,039	(14,899)	21,194	18,884	20,719	49,048	38,193	30,332	25,890	18,017	109,846
GPM (%)	3%	5%	1%	-4%	6%	6%	6%	11%	9%	7%	2%	1%	7%
qoq	-192%	78%	-72%	-396%	-242%	-11%	10%	137%	-22%	-21%			
yoy	48%	30%	-69%	37%	111%	6%	311%	-429%	80%	61%	-84%	-30%	510%
Operating Profit	(9,066)	(4,597)	(9,632)	(9,859)	(1,667)	6,156	13,265	36,062	26,443	19,290	(32,101)	(33,154)	53,816
OPM (%)	-3%	-1%	-3%	-3%	0%	2%	4%	8%	6%	5%	-2%	-2%	4%
qoq	2%	-49%	110%	2%	-83%	-469%	115%	172%	-27%	-27%			
yoy	-38%	-39%	798%	11%	-82%	-234%	-238%	-466%	-1687%	213%	-132%	3%	-262%
Domestic ASP (mn W per m ²)													
PCB	0.59	0.62	0.62	0.63	0.67	0.71	0.75	0.75	0.80	0.90	0.63	0.63	0.75
Special PCB	0.25	0.24	0.23	0.23	0.20	0.19	0.19	0.19	0.17	0.17	0.25	0.23	0.19
FPCB	0.79	0.94	0.93	0.98	0.60	0.59	0.49	0.35	0.34	0.35	0.24	0.98	0.35
Export ASP (mn W per m ²)													
PCB	1.01	1.07	1.07	1.08	1.16	1.14	1.18	1.24	1.50	1.53	1.05	1.08	1.24
Special PCB	0.12	0.13	0.13	0.13	0.14	0.14	0.13	0.14	0.18	0.18	0.13	0.13	0.14
FPCB	0.37	0.39	0.38	0.35	0.35	0.31	0.29	0.30	0.29	0.27	0.37	0.35	0.30
Order Backlog	108,719	133,743	121,390	120,870	139,730	159,968	114,941	199,155	157,254	426,639	118,628	120,870	199,155
qoq	-8%	23%	-9%	0%	16%	14%	-28%	73%	-21%	171%			
yoy	-21%	9%	2%	29%	20%	-5%	65%	13%	167%		-21%	2%	65%

Source: Company filings

Exhibit 37: TLB Financials

TLB	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	44,248	39,484	46,446	49,824	52,990	64,053	68,888	72,567	75,835	88,243	171,307	180,001	258,498
qoq	-8%	-11%	18%	7%	6%	21%	8%	5%	5%	16%			
yoy	7%	-7%	18%	4%	20%	62%	48%	46%	43%	38%	-23%	5%	44%
Revenue Breakdown													
DDR5	15,929	16,583	18,578	15,944	27,555	37,791	44,088	49,345	44,742	51,181	59,957	67,034	158,780
SSD	20,797	16,978	21,829	28,400	20,666	23,059	21,355	19,593	25,784	34,415	77,088	88,004	84,673
Others	7,522	5,923	6,038	5,481	4,769	3,203	3,444	3,628	5,308	2,647	34,261	24,963	15,044
Gross Profit	2,236	2,916	5,511	3,440	4,731	9,637	11,974	11,879	14,172	17,180	11,356	14,104	38,221
GPM (%)	5%	7%	12%	7%	9%	15%	17%	16%	19%	19%	7%	8%	15%
qoq	-36%	30%	89%	-38%	38%	104%	24%	-1%	19%	21%			
yoy	-35%	-16%	509%	-2%	112%	230%	117%	245%	200%	78%	-76%	24%	171%
Operating Profit	(261)	(59)	2,770	908	1,874	6,857	8,652	8,580	10,718	12,773	3,045	3,358	25,963
OPM (%)	-1%	0%	6%	2%	4%	11%	13%	12%	14%	14%	2%	2%	10%
qoq	-123%	-77%	-4758%	-67%	106%	266%	26%	-1%	25%	19%			
yoy	-117%	-104%	-339%	-21%	-819%	-11631%	212%	844%	472%	86%	-92%	10%	673%
Order Backlog	14,000	17,800	15,800	13,000	36,200	44,200	44,500	52,600	53,900	131,800	17,300	13,000	52,600
qoq	-19%	27%	-11%	-18%	178%	22%	1%	18%	2%	145%			
yoy	-30%	70%	-1%	-25%	159%	148%	182%	305%	49%	198%	-1%	-25%	305%
PCB ASP (W mn per m ²)	0.76	0.81	0.84	0.84	0.94	1.00	1.05	1.10	1.33	1.36	0.75	0.84	1.10
qoq	1%	7%	3%	1%	12%	6%	5%	5%	22%	2%			
yoy	1%	7%	11%	12%	24%	22%	25%	30%	42%	37%	-5%	12%	30%

Source: Company filings

3. Chip testing

In this section, we share read-across from 2Q26 earnings results from 4 major Korean chip testing companies (Leeno Industrial [Exhibit 41](#), ISC [Exhibit 42](#), Techwing [Exhibit 43](#), Doosan Tesna [Exhibit 44](#)), companies' comments, and data-points. Korean chip testing supply chain companies posted solid 2Q26 results and we believe strong demand for server CPU sockets and a positive demand outlook for memory testing sockets/equipment carry positive implications for our memory coverage.

What we learned from 2Q26 results/data-points/company comments:

#1 ISC: ISC saw strong yoy growth in 2Q26 revenue and OP (revenue +41% yoy and OP +55% yoy) driven by strong growth in memory and server CPU socket demand and resilient GPU testing socket demand. Its AI revenue increased by 71% yoy and AI revenue mix increased from 67% to 81%, implying solid growth in AI compute production <[Exhibit 39](#)>.

#2 ISC: Due to capacity tightness, customers are offering to pay higher pricing to secure production line capacity and asking for LTAs. Under current tight supply situation, the company plans to increase capacity by 2.5x by 2029.

#3 ISC: ISC mentioned testing socket demand for both compute and memory is solid and said it is particularly sensing strong demand from server CPU and HBM.

#4 ISC: Driven by solid data center demand and the transition from pogo pin sockets to rubber sockets, the company expects revenue of W310bn this year with an operating margin in the mid-30% range, followed by W440bn next year and W500bn the year after.

#5 Leeno Industrial: Leeno recorded solid revenue and OP yoy growth in 2Q26 (revenue +27% yoy, OP +38% yoy) benefiting from solid seasonal peak demand for its smartphone chip testing sockets, which consists of 60-70% of revenue. The company continues to expect solid growth in smartphone testing sockets on the back of socket ASP increases from smartphone AP spec upgrades and said it is seeing solid demand from new growth areas such as AI and servers.

#6 Techwing: Techwing saw strong profit yoy growth in 2Q26 (revenue: +16% yoy, OP:

+97%) and said orders for memory test handlers are increasing rapidly and customers are requesting shorter delivery lead times. Driven by solid order momentum, the company expects 3Q26 performance to outperform 1Q26 and 2Q26.

#7 Doosan Tesna: Doosan Tesna, which specializes on logic chip testing, made OP yoy turnaround in 2Q26. The company said it is seeing resilient demand for ADAS chip testing and said it expects revenue generation from AI chip testing to start from 4Q26. In-line with Doosan Tesna's earnings recovery, we believe Samsung Foundry's operating losses are narrowing and expect a turnaround to OP in 2H27 <Exhibit 40>.

#8 Overall, Korean chip testing companies showed a positive tone regarding market conditions especially around AI, though some expressed concern over continuing slow smartphone demand.

Implication to our coverage:

#1 Strong demand for server CPU sockets and a positive demand outlook for memory testing sockets/equipment carry positive implications for our memory coverage

#2 Doosan Tesna, which specializes on logic chip testing, made OP yoy turnaround in 2Q26 and extended earnings recovery trajectory. In-line with this trend, we believe Samsung Foundry's operating losses are narrowing and expect a turnaround to OP in 2H27

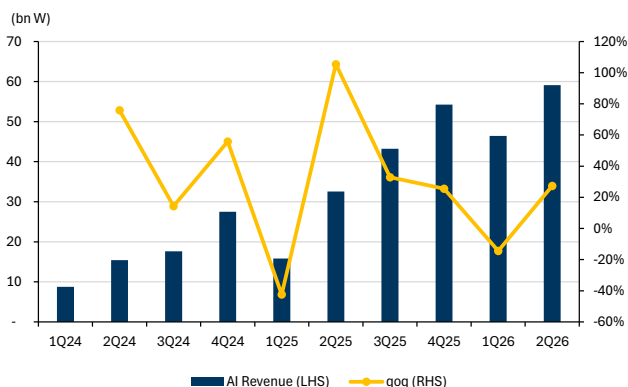
Exhibit 38: Combined earnings of 4 Korea major chip testing companies

Chip Testing	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue (W mn)	222,042	271,316	264,752	253,181	203,977	288,976	286,344	276,435	297,273	353,819	868,051	1,011,291	1,055,731
qoq	6%	22%	-2%	-4%	-19%	42%	-1%	-3%	8%	19%			
yoy	12%	13%	20%	21%	-8%	7%	8%	9%	46%	22%	-17%	17%	4%
Gross Profit (W mn)	75,642	104,110	91,530	80,576	55,909	106,157	113,417	113,396	121,652	155,310	316,078	351,857	388,878
GPM (%)	34%	38%	35%	32%	27%	37%	40%	41%	41%	44%	36%	35%	37%
qoq	2%	38%	-12%	-12%	-31%	90%	7%	0%	7%	28%			
yoy	14%	12%	11%	9%	-26%	2%	24%	41%	118%	46%	-26%	11%	11%
Operating Profit (W mn)	48,682	72,070	61,709	47,859	24,580	71,400	78,079	77,889	86,103	117,190	189,054	230,320	251,948
OPM (%)	22%	27%	23%	19%	12%	25%	27%	28%	29%	33%	22%	23%	24%
qoq	-1%	48%	-14%	-22%	-49%	190%	9%	0%	11%	36%			
yoy	24%	19%	54%	-3%	-50%	-1%	27%	63%	250%	64%	-40%	22%	9%

Revenue/GP/OP includes Leeno Industrial, ISC, Techwing, and Doosan Tesna

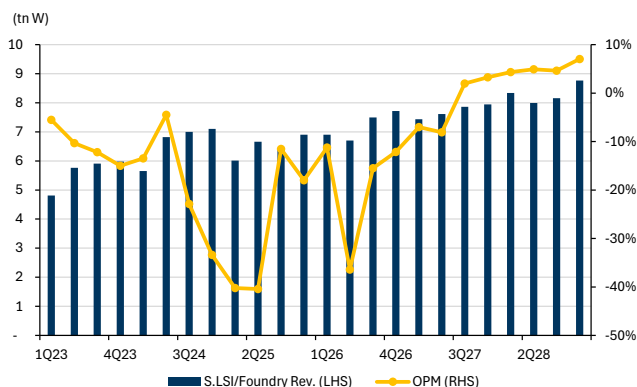
Source: Company filings

Exhibit 39: ISC's AI revenue increased by 71% yoy and AI revenue mix increased from 67% to 81%
ISC AI revenue



Source: Company filings

Exhibit 40: Samsung Foundry's operating losses are narrowing and we expect a turnaround to OP in 2H27
GSe SEC S.LSI/Foundry revenue and OPM trend and estimates



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 41: Leeno Industrial Financials

Leeno Industrial	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	54,852	70,979	68,936	83,419	78,410	112,522	96,842	84,760	99,771	143,187	255,573	278,186	372,534
qoq	-5%	29%	-3%	21%	-6%	44%	-14%	-12%	18%	44%	-21%	9%	34%
yoy	12%	-6%	-6%	44%	43%	59%	40%	2%	27%	27%	-21%	9%	34%
Revenue Breakdown											255,573	278,186	372,534
Leeno Pin/Socket	47,124	63,663	61,837	74,519	68,367	101,742	87,458	73,615	88,579	134,349	228,262	247,142	331,183
Medical Device	7,728	7,316	7,099	8,900	10,043	10,780	9,383	11,145	11,192	8,838	27,311	31,044	41,351
Gross Profit	26,322	36,849	33,872	41,123	38,755	58,877	52,734	44,034	51,494	78,876	128,335	138,166	194,400
GPM (%)	48%	52%	49%	49%	49%	52%	54%	52%	52%	55%	50%	50%	52%
qoq	-20%	40%	-8%	21%	-6%	52%	-10%	-16%	17%	53%	-17%	8%	41%
yoy	35%	-4%	-10%	25%	47%	60%	56%	7%	33%	34%	-17%	8%	41%
Operating Profit	23,302	33,210	30,669	37,020	34,938	53,444	48,261	40,354	47,301	73,514	114,378	124,201	176,997
OPM (%)	42%	47%	44%	44%	45%	47%	50%	48%	47%	51%	45%	45%	48%
qoq	-23%	43%	-8%	21%	-6%	53%	-10%	-16%	17%	55%	-16%	9%	43%
yoy	35%	-1%	-8%	22%	50%	61%	57%	9%	35%	38%	-16%	9%	43%

Source: Company filings

Exhibit 42: ISC Financials

ISC	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	35,073	49,758	50,393	39,257	31,683	51,659	64,522	72,330	68,272	72,929	140,222	174,480	220,195
qoq	41%	42%	1%	-22%	-19%	63%	25%	12%	-6%	7%	-22%	24%	26%
yoy	-24%	38%	53%	57%	-10%	4%	28%	84%	115%	41%	-22%	24%	26%
Revenue Breakdown													
Datacenter	17,536	19,903	22,677	27,872	16,792	33,062	43,875	54,248	45,742	54,697	63,100	87,988	147,977
Smartphone	10,522	11,942	15,118	6,281	9,505	13,948	13,550	12,296	15,020	14,586	42,067	43,863	49,299
PC/Notebook	2,806	7,961	10,079	3,533	3,168	3,100	4,517	2,893	4,096	2,188	28,044	24,379	13,678
Auto	1,754	1,990	1,512	1,178	951	517	1,290	2,170	2,048	1,459	4,207	6,433	4,927
Wearable	1,754	2,986	1,008	-	951	517	645	723	683	-	2,804	5,747	2,836
Others	351	4,976	504	393	317	517	645	-	683	-	1,402	6,223	1,479
Gross Profit	14,933	22,348	21,201	15,916	14,239	22,549	27,605	33,346	35,386	34,291	55,165	74,398	97,739
GPM (%)	43%	45%	42%	41%	45%	44%	43%	46%	52%	47%	39%	43%	44%
qoq	69%	50%	-5%	-25%	-11%	58%	22%	21%	6%	-3%	-35%	35%	31%
yoy	-22%	51%	70%	80%	-5%	1%	30%	110%	149%	52%	-35%	35%	31%
Operating Profit	8,558	14,912	13,814	7,500	6,992	13,713	17,421	21,929	23,616	21,320	10,737	44,784	60,054
OPM (%)	24%	30%	27%	19%	22%	27%	27%	30%	35%	29%	8%	26%	27%
qoq	247%	74%	-7%	-46%	-7%	96%	27%	26%	8%	-10%	-81%	317%	34%
yoy	-13%	130%	-273%	204%	-18%	-8%	26%	192%	238%	55%	-81%	317%	34%
AI Revenue	8,768	15,425	17,637	27,480	15,842	32,545	43,230	54,248	46,425	59,072	5,679	69,310	145,864
Gen-AI	5,875	10,026	11,464	25,281	11,089	30,918	38,042	51,535	41,783	53,756	3,691	52,646	131,585
On-Device AI	2,893	5,399	6,173	2,198	4,753	1,627	5,188	2,712	4,643	5,317	1,988	16,664	14,280
Non-AI Revenue	26,304	34,333	32,755	11,777	15,842	19,114	21,292	18,083	21,847	13,857	57,421	105,170	74,330

Source: Company filings

Exhibit 43: Techwing Financials

Techwing	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	40,339	50,770	46,722	47,676	34,587	48,894	41,766	33,882	52,400	56,777	133,604	185,507	159,128
qoq	-2%	26%	-8%	2%	-27%	41%	-15%	-19%	55%	8%	-50%	39%	-14%
yoy	41%	29%	92%	16%	-14%	-4%	-11%	-29%	52%	16%	-50%	39%	-14%
Revenue Breakdown													
Semiconductor	34,217	45,323	44,768	39,596	31,679	45,253	41,375	29,846	47,948	54,655	114,386	163,904	148,153
Display	6,123	5,446	1,954	8,080	2,908	3,641	390	4,037	4,452	2,096	18,630	21,603	10,976
Others	-	-	-	-	-	-	-	-	-	25	588	-	-
Gross Profit	16,041	20,186	18,735	16,524	14,645	18,829	19,359	14,362	21,537	25,179	49,548	71,486	67,195
GPM (%)	40%	40%	40%	35%	42%	39%	46%	42%	41%	44%	37%	39%	42%
qoq	24%	26%	-7%	-12%	-11%	29%	3%	-26%	50%	17%	-55%	44%	-6%
yoy	33%	30%	107%	27%	-9%	-7%	3%	-13%	47%	34%	-55%	44%	-6%
Operating Profit	5,573	7,175	6,177	4,486	1,783	6,361	7,111	582	9,702	12,505	3,160	23,411	15,838
OPM (%)	14%	14%	13%	9%	5%	13%	17%	2%	19%	22%	2%	13%	10%
qoq	84%	29%	-14%	-27%	-60%	257%	12%	-92%	1567%	29%	-95%	641%	-32%
yoy	1324%	195%	-330%	48%	-68%	-11%	15%	-87%	444%	97%	-95%	641%	-32%
Order Backlog Breakdown	28,468	28,449	19,902	26,792	19,416	10,572	9,435	35,940	48,576	60,268	32,640	26,792	35,940
Semiconductor	21,165	19,835	14,234	22,238	18,425	9,652	5,664	32,252	47,055	55,508	26,527	22,238	32,252
Display	7,303	8,614	5,668	4,554	991	920	3,771	3,688	1,521	4,760	6,113	4,554	3,688

Source: Company filings

Exhibit 44: Doosan Tesna Financials

Doosan Tesna	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	91,777	99,809	98,702	82,829	59,296	75,901	83,214	85,463	76,829	80,926	338,652	373,118	303,874
qoq	7%	9%	-1%	-16%	-28%	28%	10%	3%	-10%	5%			
yoy	23%	12%	11%	-3%	-35%	-24%	-16%	3%	30%	7%	22%	10%	-19%
Revenue Breakdown													
Wafer Test	83,553	93,525	91,572	76,822	54,455	70,117	77,598	78,795	71,334	73,271	323,624	345,472	280,964
PKG Test	4,909	4,438	4,600	4,315	3,625	4,455	4,394	5,549	4,335	7,035	15,028	18,262	18,023
Die Preperation	3,315	1,846	2,530	1,693	1,216	1,329	1,222	1,120	1,161	620	0	9,384	4,887
Gross Profit	18,346	24,728	17,721	7,012	(11,730)	5,902	13,718	21,653	13,235	16,964	83,031	67,808	29,544
GPM (%)	20%	25%	18%	8%	-20%	8%	16%	25%	17%	21%	25%	18%	10%
qoq	-5%	35%	-28%	-60%	-267%	-150%	132%	58%	-39%	28%			
yoy	15%	3%	-25%	-64%	-164%	-76%	-23%	209%	-213%	187%	3%	-18%	-56%
Operating Profit	11,249	16,773	11,049	(1,147)	(19,133)	(2,118)	5,285	15,025	5,484	9,851	60,779	37,923	(941)
OPM (%)	12%	17%	11%	-1%	-32%	-3%	6%	18%	7%	12%	18%	10%	0%
qoq	-16%	49%	-34%	-110%	1568%	-89%	-350%	184%	-64%	80%			
yoy	-5%	-8%	-36%	-109%	-270%	-113%	-52%	-1410%	-129%	-565%	-10%	-38%	-102%
Utilization Rate (Cumulative)													
Wafer Test	62.9%	65.3%	64.9%	61.9%	35.6%	40.1%	43.2%	45.5%	53.4%	57.8%	69.7%	61.9%	45.5%
PKG Test	45.3%	43.2%	49.9%	53.8%	62.2%	67.5%	62.4%	58.9%	44.3%	58.2%	34.4%	53.8%	58.9%
Die Preperation				25.4%	11.8%	12.7%	12.5%	12.2%	11.6%	8.9%	36.2%	25.4%	12.2%

Source: Company filings

4. HBM

In this section, we share read-across from 2Q26 earnings results from Hanmi Semiconductor <Exhibit 45>, companies' comments, and data-points. Solid 2Q26 earnings, optimistic guidance, and upbeat commentary from HBM supply chain companies carry positive implications for our memory coverage. In line with recent data-points and comments, we estimate the HBM TAM to grow by 67% and 108% yoy this year and next year, respectively.

What we learned from 2Q26 results/data-points/company comments:

#1 Hanmi Semiconductor posted record-high quarterly results in 2Q26. During the recent press interview with the CEO <link>, the CEO highlighted potential TC bonder supply shortages due to significant HBM capex by memory makers, accelerated fab ramp-ups, and increasing spending from foundry companies. In response, Hanmi Semiconductor said plans to complete its W100bn invested hybrid bonder factory (Plant 7) by 1H27 and is reportedly acquiring existing plant for its Plant 8 <link>.

#2 ISC expects its HBM test socket revenue to more than double this year. Techwing also anticipates a substantial surge in orders for its HBM cube prober in the second half of this year and into next year, driven by expanding volume from existing clients and the addition of a new customer.

#3 We track 1) import data of epoxide resins from Japan to Icheon/Cheongju (where Hynix HBM production bases are located) as, based on industry checks/media reports, we believe Hynix mainly sources its MR-MUF materials from Japanese company Namics, and 2) import data of plastic film from Japan to Hwaseong/Pyeongtaek (where SEC HBM production bases are located) as, based on our checks, we believe SEC mainly sources its TC-NCF materials from Japanese company Resonac. Import trends of epoxide resins/plastic film have been showing strong correlation with Hynix/SEC HBM shipment (Hynix: Exhibit 46/ SEC: Exhibit 47), respectively; hence we believe these could be good proxies to gauge HBM shipment scale of Hynix/SEC.

In July, import value of epoxide resins from Japan to Icheon/Cheongju decreased by 18% yoy and import value of plastic film from Japan to Hwaseong/Pyeongtaek increased by 134% yoy. YTD July, import value of epoxide resins from Japan to Icheon/Cheongju increased by 1% yoy and import value of plastic film from Japan to Hwaseong/Pyeongtaek increased by 84% yoy, likely implying SEC's solid HBM

production ramp-up.

Implication to our coverage:

#1 Solid 2Q26 earnings, optimistic guidance, and upbeat commentary from HBM supply chain companies carry positive implications for our memory coverage. In line with recent data-points and comments, we estimate the HBM TAM to grow by 67% and 108% yoy this year and next year, respectively <Exhibit 50>.

#2 SEC plastic film imports for HBM increased 134% yoy in July likely implying SEC's solid HBM production ramp-up. SEC guided its HBM4 revenue will more than triple qoq in 3Q26 and HBM4 revenue mix will exceed 60% of HBM revenue in 2H26. Reflecting strong HBM4 guidance, we expect SEC HBM revenue to grow 234% yoy this year <Exhibit 48> and recover market share <Exhibit 48>.

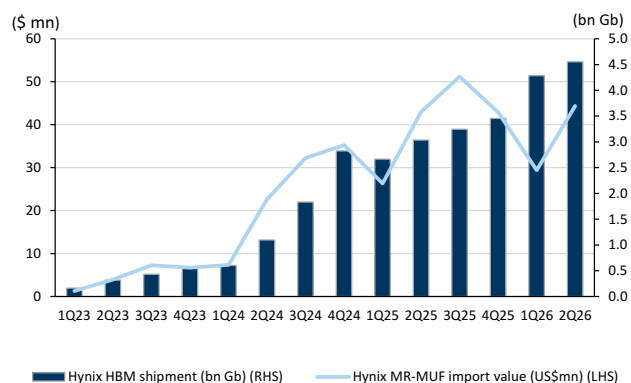
Exhibit 45: Hanmi Semiconductor Financials

Hanmi Semiconductor	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	2023	2024	2025
Revenue	77,318	123,478	208,525	149,596	147,392	180,047	166,231	83,014	50,902	251,163	159,009	558,917	576,685
qoq	48%	60%	69%	-28%	-1%	22%	-8%	-50%	-39%	393%			
yoy	192%	152%	568%	186%	91%	46%	-20%	-45%	-65%	39%	-51%	252%	3%
Revenue Breakdown													
Export	20,124	24,807	95,751	89,957	129,870	140,086	129,669	65,674	45,772	182,760	116,763	230,639	465,299
Import	57,194	98,671	112,774	59,639	17,522	39,961	36,563	17,340	5,130	68,403	42,246	328,278	111,386
Gross Profit	38,931	68,053	120,400	87,138	92,307	106,493	86,670	46,598	27,559	155,362	79,404	314,522	332,068
GPM (%)	50%	55%	58%	58%	63%	59%	52%	56%	54%	62%	50%	56%	58%
qoq	28%	75%	77%	-28%	6%	15%	-19%	-46%	-41%	464%			
yoy	250%	172%	842%	186%	137%	56%	-28%	-47%	-70%	46%	-57%	296%	6%
Operating Profit	28,706	55,438	99,305	71,943	69,632	86,308	67,822	27,628	8,456	130,347	34,571	255,392	251,390
OPM (%)	37%	45%	48%	48%	47%	48%	41%	33%	17%	52%	22%	46%	44%
qoq	56%	93%	79%	-28%	-3%	24%	-21%	-59%	-69%	1441%			
yoy	1283%	396%	3321%	291%	143%	56%	-32%	-62%	-88%	51%	-69%	639%	-2%

Source: Company filings

Exhibit 46: Import trends of epoxide resins have been showing strong correlation with Hynix HBM shipments

Import trend of epoxide resins from Japan to Hynix

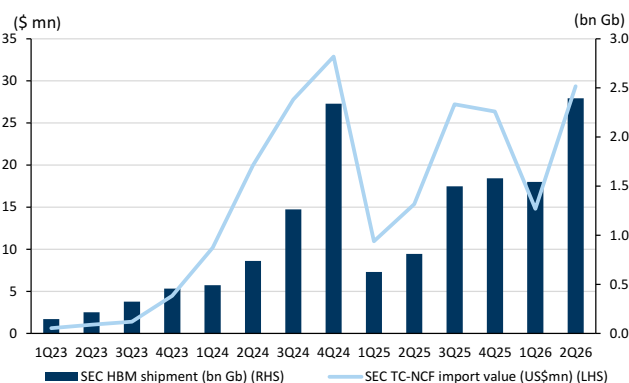


Epoxide resins import (JP to Hynix) = Import value of epoxide resins from Japan to Icheon/Cheongju where Hynix's HBM production bases are located.

Source: MOTIE, TRASS, Goldman Sachs Global Investment Research

Exhibit 47: Import trend of plastic film has been showing strong correlation with SEC HBM shipment

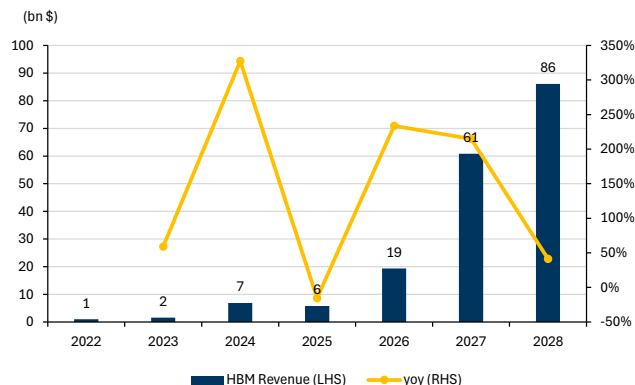
Import trend of plastic film from Japan to SEC



Plastic film import (JP to SEC) = Import value of plastic film from Japan to Hwaseong/Pyeongtaek where SEC HBM production bases are located.

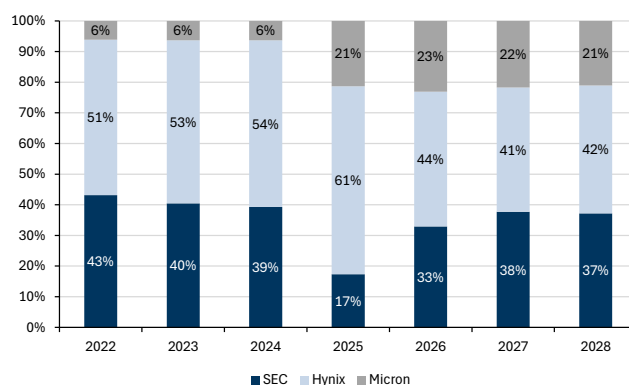
Source: MOTIE, TRASS, Goldman Sachs Global Investment Research

Exhibit 48: SEC HBM revenue and growth trend



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 49: HBM market share



Data for 2022, 2023, 2024, and 2025 are estimated

Source: Goldman Sachs Global Investment Research

Exhibit 50: HBM demand and TAM

	2022	2023	2024	2025	2026E	2027E	2028E
Total HBM demand ('000 GB)	215,999	349,471	1,492,389	2,677,990	4,724,443	6,840,511	9,712,529
yoy%		62%	327%	79%	76%	45%	42%
% of total DRAM demand	1%	1%	5%	7%	10%	12%	14%
GPU portion of HBM demand		80%	75%	79%	67%	64%	61%
ASIC portion of HBM demand		20%	25%	21%	33%	36%	39%
HBM demand from GPU ('000 GB)	280,743	1,123,471	2,103,438	3,162,046	4,351,337	5,911,903	
GPU HBM demand yoy%			300%	87%	50%	38%	36%
HBM demand from ASIC ('000 GB)	68,728	368,918	574,552	1,562,396	2,489,174	3,800,626	
ASIC HBM demand yoy%			437%	56%	172%	59%	53%
Average HBM ASP (US/GB)	10.8	11.3	11.7	12.5	11.8	17.0	17.3
yoy change		5%	3%	7%	-6%	44%	2%
HBM TAM (mn)	2,333	3,963	17,437	33,449	55,764	116,267	168,384
yoy change		70%	340%	92%	67%	108%	45%

Data for 2022, 2023, 2024, and 2025 are estimated

Source: Goldman Sachs Global Investment Research

Exhibit 51: Peer Comps

Semi-Supply Chain Company	Market Cap (bn KRW)	Avg. daily value traded	P/E (x)			P/B (x)			EV/EBITDA (x)			ROE (%)			OPM (%)		
			2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E
Front-end equipment																	
Jusung Engineering	7,976	252	115.2	52.0	20.6	11.9	9.7	6.6	94.1	40.0	16.1	9%	19%	38%	23%	28%	30%
Wonik IPS	5,365	133	24.9	22.1	17.2	4.6	3.8	3.2	22.7	14.9	11.5	19%	19%	20%	15%	18%	19%
EO Technics	5,267	53	43.1	31.8	26.6	6.8	5.7	4.8	32.4	22.9	18.4	17%	19%	19%	27%	30%	31%
PSK	3,835	95	24.8	19.2	15.1	5.7	4.5	3.5	17.9	13.2	10.2	25%	26%	26%	27%	28%	28%
HPSP	3,761	102															
Eugene Tech	3,105	33	25.3	21.9	16.9	5.4	4.4	3.6	24.5	14.9	11.2	22%	21%	22%	20%	25%	26%
Tes	2,852	127	26.8	21.5	17.3	5.6	4.6	3.7	23.8	17.6	13.5	24%	24%	24%	22%	23%	25%
KC Tech	1,350	31	20.3	16.6	14.9	2.4	2.1	1.9				17%	18%	13%	19%	20%	21%
SEMES																	
Substrate / CCL																	
Doosan	19,676	113	39.4	22.9	19.2	10.1	7.7	6.2	13.0	9.9	7.8	26%	29%	26%	8%	10%	12%
Isu Petasys	7,657	104	27.6	17.7	13.9	7.5	5.3	3.9	19.2	12.3	9.3	31%	36%	32%	21%	24%	25%
Daeduck Electronics	5,643	100	25.8	17.7	14.4	5.3	4.2	3.3	14.6	10.2	8.0	22%	26%	25%	17%	20%	21%
Simmtech	4,241	165	22.3	13.0	9.6	5.6	3.9	2.8	14.1	8.6	6.4	28%	35%	33%	12%	15%	19%
Korea Circuit	1,511	42	12.1	8.0	5.9	3.0	2.3	1.7				25%	28%	29%	8%	11%	13%
TLB	826	26	19.5	14.2	9.0	3.2	2.9	2.2	12.2	8.3	5.1	24%	24%	29%	15%	18%	20%
Chip testing																	
Leeno Industrial	5,274	63															
ISC	3,796	34	41.0	29.2	22.5	6.1	5.2	4.4	29.6	20.4	15.3	16%	19%	21%	33%	36%	37%
Techwing	1,810	36	27.6	8.6	8.3				16.0	8.1	6.9	28%	44%	38%	27%	34%	34%
Doosan Tesna	1,546	39															
HBM																	
Hanmi Semiconductor	21,493	327	69.2	44.7	37.8	24.5	17.6	14.7	56.0	34.8	29.2	41%	44%	39%	48%	51%	58%

Average daily value traded is based on 1 month (20 trading days), data as of 2026.08.20

Source: Bloomberg

Price Target Risks and Methodology - Samsung Electronics

Valuation methodology: Our 12m 2026-2027E EV/EBITDA-based SOTP target price for

the common share is W490,000. Our 12-month target price for the preference share is W360,000, which is based on our target preferred-to-common shares discount of 27%, derived from averaging: 1) the preferred discount of the 2-factor model and 2) the average preference share discount to common shares during the past 1 month. We are Buy rated on both the common and preference shares.

Key downside risks: 1) major deterioration in memory supply/demand, 2) sharp contraction in smartphone margins, and 3) mobile OLED market share loss.

Price Target Risks and Methodology - SK Hynix Inc.

Valuation methodology: Our 2026E/27E avg. P/E-based 12m TP is W3,500,000, applying a target P/E multiple of 9.0X.

Key risks: Key risks include 1) major deterioration in memory supply/demand and delay in technology migration, 2) weaker demand for smartphones/PCs/servers which would impact overall conventional memory demand, 3) Samsung's positive HBM business progress which would impact HBM revenue and profit, 4) lower AI-related capex which would impact overall HBM demand, and thus HBM revenue/profit for the company.

Price Target Risks and Methodology - Samsung Electro-Mechanics

Valuation: We are Buy rated on Samsung Electro-Mechanics with a SOTP-based (2027E-2028E EV/EBITDA driven) 12-month TP of W2,250,000.

Key downside risks: A stronger-than-expected industry MLCC supply increase and weaker-than-expected demand for AI server/auto/smartphones.

Disclosure Appendix

Reg AC

We, Giuni Lee, Taeyong Lee and Daiki Takayama, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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